Leps-P: Probabilistic Strength Envelopes with Dilatancy-Bounded Curvature for Rockfill and Mine Waste Rock

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Abstract

Reliability-based design of high waste dumps needs the drained shear strength of rockfill and mine waste rock as a distribution, together with its dependence on stress and a rule to combine published experience with site tests. Leps-P is a three-level hierarchical model of the secant friction angle in which the dilatant part of strength decays with stress toward an asymptote, so that the curvature of the envelope decreases as dilatancy is suppressed. It was calibrated on 767 large-scale triaxial tests from 36 publications in the public Leps 2.0 database and evaluated by leaving out one publication at a time. The data supported a curvature that decays with stress (likelihood ratio test, $p < 0.001$) but did not identify the asymptotic angle, which was therefore averaged over the likelihood ridge. Out of sample, the bounded form scored better than a quadratic log–log form (continuous ranked probability score 0.0670 versus 0.0688, Wilcoxon $p = 0.04$) and removed the low bias of the median above 2.75 MPa. With a few site tests at low stress, a prespecified level-only update predicted the strength at higher stress better than a least-squares fit with a fixed curvature (score 0.0278 versus 0.0369). At field scale, the uncertainty of an untested waste rock came mainly from specimen size and from material and laboratory variability; the unidentified asymptote contributed about 4% of the variance at 10 MPa. Within-material comparisons did not detect a systematic specimen-size effect. For a 700 m dump of waste rock from an open-pit mine, placement density and untested rock types controlled the margin of safety. The model, design charts, and code allow re-estimation as tests at higher stress become available.

Practical Applications

Designers of high waste dumps and rockfill dams often have few or no large-scale shear tests of their own and rely on published charts of rockfill strength. This work turns the most widely used chart into a tool that gives a best estimate of strength together with a cautious value for design, for any rock quality and density. It shows that the strength of crushed rock flattens at high pressure in the way expected from the loss of dilation, and that this flattening can be used safely within the range of available tests. A few laboratory tests of the site material at low pressure sharpen the estimate at higher pressure, and the tool warns the user when site tests disagree with published experience. Most of the remaining uncertainty for an untested material comes from the size of the rock particles and from differences between materials and laboratories, so testing the actual waste rock at large scale is the most valuable investigation. For a very high dump, the loose state of placed waste rock and rock types without tests governed the safety margin. Free software accompanies the paper.

Author keywords: Rockfill; Mine waste rock; Shear strength; Nonlinear failure envelope; Dilatancy; Hierarchical model; Reliability-based design; High waste dumps.

Introduction

Leps (1970) compiled 109 large-scale triaxial tests and proposed three nonlinear envelopes for rockfill that have been used for dams and waste dumps whenever site-specific tests were unavailable. Cylwik et al. (2025) updated them with a public database of 938 tests (Arrieta 2025), adopted the power criterion of De Mello (1977) and Charles and Watts (1980) with a single fitting parameter, the secant friction angle at $\sigma_n' = 10p_a$, and related that angle to rock quality and void ratio. That recent and widely adopted update is referred to as Leps 2.0.

Two developments motivate a further extension. First, waste dumps now reach 500–700 m (Valenzuela et al. 2008), so effective normal stresses on deep failure surfaces approach or exceed the 7.9 MPa reached in the database, in a material that crushes and loses dilatancy (Marsal 1973; Hardin 1985; Lade et al. 1996; Linero et al. 2007; Ovalle et al. 2020). How the envelope behaves beyond the data then matters, and a power law with constant exponent keeps curving downward indefinitely, whereas the strength of a granular material is bounded below by its critical state (Bolton 1986; Jefferies and Been 2016). Second, design is moving toward reliability-based criteria (Read and Stacey 2009; Hawley and Cunning 2017), which require a strength distribution and a rule to combine published experience with site tests.

Hierarchical models that borrow strength from generic databases have been developed for soil properties (Ching and Phoon 2019; Ching et al. 2021), and a hierarchical Bayesian analysis of the Barton and Kjærnsli (1981) parameters has been applied to rockfill groups of a single project (Contreras et al. 2022; Contreras and Linero-Molina 2025). The contributions of this work are (1) an envelope whose curvature decays with stress as dilatancy is suppressed, tested against a fixed-form alternative out of sample and within materials; (2) the separation of publication and material variability; (3) a prespecified site-updating rule with a prior–data conflict check, compared with the simple fits used in practice; (4) a decomposition of the uncertainty of an untested waste rock at field scale; and (5) design charts and an open implementation. Leps-P is an independent extension built on the public Leps 2.0 database; it is not affiliated with the authors of Leps 2.0.

Database and Model Formulation

Calibration Database

The public database (Arrieta 2025) was filtered to tests with $\sigma_3' \geq 60$ kPa, reported initial void ratio, and at least two tests per set, giving 767 tests on 226 sets from 36 publications (Fig. 1). A set is a material with a given gradation, density, and rock type tested in one program; failure is the maximum deviator stress, as in Leps 2.0. Rock quality classes follow Table 5 of Cylwik et al. (2025). Tests with $\sigma_n' > 3$ MPa (90 tests) come from eight publications, seven of them published between 1963 and 1983 (e.g., Marachi et al. 1972; Marsal 1973), and sets with $\phi_1' < 31°$ were excluded by Cylwik et al. (2025), so the lower tail of the population is truncated.

Mechanical Basis of the Envelope

The Leps 2.0 criterion is the power law

$$\tau = \tan\phi_1' \sigma_0^{1-m} {\sigma_n'}^m \tag{1}$$

where $\sigma_0 = 10p_a$, $p_a = 0.1013$ MPa, ϕ_1' is the secant friction angle at $\sigma_n' = \sigma_0$, and m is a constant exponent; the secant friction angle is $\phi_s' = \arctan(\tau/\sigma_n')$. In granular materials the excess of strength over the critical-state value is the contribution of dilatancy, which is governed by the state of the material relative to its critical state and is suppressed by high confining stress (Bolton 1986; Jefferies and Been 2016, p. 43); particle crushing reinforces this trend (Hardin 1985; McDowell and Bolton 1998; Einav 2007). A minimal form with this property is

$$\tan\phi_s' = \tan\phi_{cs}' + (\tan\phi_1' - \tan\phi_{cs}')\exp(-\zeta x), x = \ln(\sigma_n'/\sigma_0) \tag{2}$$

in which the dilatant part decays at rate ζ and ϕ_s' tends to an asymptotic angle ϕ_{cs}'. For $\phi_{cs}' = 0$ it reduces to a power law with $m = 1 - \zeta$. Because ψ-type state measures (Jefferies and Been 2016) combine void ratio and stress, the effect of void ratio enters through $\tan\phi_1'$ and therefore fades

with stress, as observed in large triaxial tests (Marachi et al. 1972; Xiao et al. 2014a). Here ϕ'_{cs} is a population asymptote, not a measured critical-state angle.

Hierarchical Model

With $y = \ln(\tau/\sigma_0)$, each set j of publication k is a realization of a population:

$$y_{ijk} = x_{ijk} + \ln\left[T + (e^{\eta_{jk}} - T)e^{-\zeta x_{ijk}}\right] + b_{0,jk} + b_{1,jk}x_{ijk} + \varepsilon_{ijk}, \eta_{jk} = \beta_{c(jk)} + \beta_e e_{jk} \quad (3)$$

$$(b_0, b_1)_{jk} = \mathbf{b}_k^{(p)} + \mathbf{b}_{jk}^{(s)}, \mathbf{b}^{(p)} \sim N(\mathbf{0}, \mathbf{G}_p), \mathbf{b}^{(s)} \sim N(\mathbf{0}, \mathbf{G}_s), \ln\sigma_\varepsilon(x) = a_0 + a_1 x \quad (4)$$

where T = tan ϕ'_{cs}, β_c is the coefficient of rock quality class c, β_e the effect of initial void ratio e, $\mathbf{b}^{(p)}$ and $\mathbf{b}^{(s)}$ the publication and set random effects on the level and slope of the log–log envelope with covariance matrices $\mathbf{G}_p$ and $\mathbf{G}_s$, and ε the dispersion of tests on one material, with standard deviation σ_ε (Laird and Ware 1982). The publication effect combines laboratory, procedure, era, and material selection, which cannot be separated with these data; it plays the role of measurement and transformation uncertainty in the sense of Phoon and Kulhawy (1999), but because true strength and laboratory bias cannot be distinguished it is conservatively treated as uncertainty of strength. Through β_c and β_e the median ϕ'_1 reproduces the relation $\phi'_1 = \phi'_{e=0} -$ ωe of Leps 2.0. A quadratic log–log form, $y = \theta + mx + c_2x^2$, and a power form ($c_2 = 0$), with the same random effects, were fitted for comparison.

Non-Identified Asymptote

The likelihood of (ϕ'_{cs}, ζ) has a ridge: a lower asymptote with a slower decay fits the data almost as well as a higher asymptote with a faster decay, because both shapes coincide within the stress range of the tests. Rather than fixing ϕ'_{cs}, the profile likelihood was evaluated on a grid (12–32° and 0.15–0.45), the grid points holding 99.5% of the normalized likelihood (flat prior on the grid) were refitted, and predictions average over them with these weights.

Predictive Distribution

For a new material of class *c* and void ratio *e* tested by a new laboratory, *y* is approximately normal with mean $f(x; \widehat{\boldsymbol{\beta}})$ given by the first two terms of Eq. (3) and variance

$$\mathrm{Var}[y] = \mathbf{z}^T(\mathbf{G}_p + \mathbf{G}_s)\mathbf{z} + \mathbf{j}^T\mathbf{C}_\beta\mathbf{j} + \sigma_\varepsilon^2(x), \mathbf{z} = (1, x)^T \tag{5}$$

where $*j$ *is the gradient of* f* with respect to the fixed effects and $\mathbf{C}_\beta$ their covariance. The first two terms describe the uncertainty of the mean envelope of the material, which does not average out along a failure surface within that material (Christian et al. 1994); the last describes test-to-test scatter. In the implementation, variance parameters are sampled from their posterior, (ϕ'_{cs}, ζ) from the ridge weights, and fixed effects from N($\widehat{\boldsymbol{\beta}}$, $\mathbf{C}_\beta$); samples whose local slope exceeds one for $\sigma'_n \leq 20$ MPa are discarded, because a convex envelope has no support in granular materials. Fig. 2 and Table 3 give the resulting median and 5th-percentile envelopes for untested materials.

Updating with Site Tests

Site triaxial tests are converted to (σ'_n, τ) with Eqs. (13), (14), and (16) of Cylwik et al. (2025), using the same failure criterion as the database. Consistency with the prior is checked with the prior predictive Mahalanobis distance

$$D^2 = \mathbf{r}^T\left(\mathbf{Z}(\mathbf{G}_p + \mathbf{G}_s)\mathbf{Z}^T + \mathbf{J}\mathbf{C}_\beta\mathbf{J}^T + \mathbf{S}\right)^{-1}\mathbf{r} \sim \chi^2_n \tag{6}$$

where $\mathbf{r}$ holds the residuals r_i of the *n* site tests from the prior mean, $\mathbf{Z}$ has rows (1, x_i), and $\mathbf{S} = \mathrm{diag}\sigma_\varepsilon^2(x_i)$. A *p*-value below 0.05 flags the tests for review. The prespecified rule updates only the level of the envelope at the centroid $\overline{x}$ of the tests: with $q_0 = b_0 + b_1\overline{x}$ and prior variance $g_0 = (1, \overline{x})(\mathbf{G}_p + \mathbf{G}_s)(1, \overline{x})^T$, the posterior of q_0 is normal with

$$\hat{q}_0 = \frac{\sum_i r_i/\sigma^2_{\varepsilon,i}}{1/g_0 + \sum_i 1/\sigma^2_{\varepsilon,i}}, \mathrm{Var}(q_0) = \left(\frac{1}{g_0} + \sum_i \frac{1}{\sigma^2_{\varepsilon,i}}\right)^{-1} \tag{7}$$

and the slope keeps its prior distribution conditional on q_0, so the envelope pivots about the tested range. Eq. (7) is a cut (modular) update, which treats each residual as a measurement of the level at $\overline{x}$; it can be applied in a spreadsheet. When several materials of one site were tested by the same laboratory, their publication-level effect is shared and all sets are updated jointly.

Placement Density and Specimen Size

A change of void ratio Δe from the specimens to the placed material enters through η (β_e Δe), so its effect decreases with stress. Specimen size enters as a shift of level

$$\Delta y = \beta_D \ln\left(D_f / D_l\right) \tag{8}$$

where D_f and D_l are the maximum particle sizes in the field and in the laboratory. Weibull scaling predicts $\beta_D < 0$ (Frossard et al. 2012; Ovalle et al. 2014; Alonso et al. 2012). β_D was estimated within materials of the database tested at several sizes.

Estimation and Out-of-Sample Evaluation

Fixed effects were estimated by Gauss–Newton iteration on the linearized mean with restricted maximum likelihood (REML) for the variance parameters; random effects were integrated analytically. Variance parameters were then sampled with an adaptive Metropolis algorithm under weakly informative priors (half-normal with scale 0.25 on standard deviations, uniform on correlations, normal on a_0, a_1), with four chains started at dispersed values; the fixed effects are thus estimated conditionally (an empirical Bayes scheme). Convergence was checked with the rank-normalized potential scale reduction factor and bulk and tail effective sample sizes (Vehtari et al. 2021).

Each publication was left out in turn. In each fold, (ϕ'_{cs}, ζ) were selected on the training data only and the model was refitted. In Mode A (no site tests), every test of the held-out publication was predicted. In Mode B, each held-out set with at least two tests at $\sigma'_n \leq 1.5$ MPa

and one at $\sigma_n' \geq 2$ MPa was updated with the former and the latter were predicted. The primary Mode B procedure (level-only update of every set) and the secondary comparisons were fixed before the bounded form was evaluated: the prior, a joint update of level and slope, the level-only update after discarding flagged sets, and least-squares fits of ϕ_1' with $m = 0.90$ or with m equal to the median within-set slope of the training data; the predictive standard deviation of the fits was their out-of-sample error on the training publications. Scores were the coverage of the 80% interval, the fraction of tests below the 10th percentile, and the continuous ranked probability score (CRPS) of ln τ (Gneiting and Raftery 2007). Differences were assessed by bootstrapping publications and by the Wilcoxon signed-rank test on publication means (Wilcoxon 1945). The Mode A reference was the published Leps 2.0 envelope with a constant dispersion fitted to the training data; because its constants were fitted to the whole database, this comparison favors the reference. A within-material shape test repeated Mode B for the quadratic and bounded forms on the same sets. The folds used plug-in predictive distributions (point estimates of all parameters); with all data, the distribution of the implementation, which also propagates the posterior of the variance parameters, the ridge, and the slope restriction, differed from the plug-in distribution by at most 0.026 in median ln τ and had a standard deviation 0.94–1.09 times as large, so the evaluation applies to the implementation.

Results

Parameters

Table 1 lists the parameters. The rank-normalized potential scale reduction factor was at most 1.013, and the bulk and tail effective sample sizes were at least 513 and 938. Replacing the half-normal scale of 0.25 by 0.1 or 0.5 changed the median between-publication standard deviation of the level from 0.060 to 0.060 and 0.061. At σ_0, publication effects account for 36%

(90% credible interval 20%–52%) of the between-set and between-publication variance of the level.

Evidence for Stress-Dependent Curvature

Setting ϕ'_{cs} = 0, which leaves a power law in tan ϕ'_s with constant exponent, reduced the maximum log-likelihood by 7.7 (likelihood ratio test with one degree of freedom, p = 8.5e-05). The bounded form exceeded the quadratic form, which has the same number of parameters, by 5.3 in log-likelihood. Within 3 log-likelihood units of the maximum, ϕ'_{cs} ranged from 13° to 28° (Fig. 3a); the weighted means were ϕ'_{cs} = 22.4° and ζ = 0.26. In the within-material shape test (64 tests, 46 sets, 13 publications), the bounded form reduced the mean bias of ln τ at higher stress from +0.0134 to +0.0029 and the CRPS from 0.0298 to 0.0278, and scored better in 10 of 13 publications (Wilcoxon p = 0.027; Fig. 3b). For the site tests of the application, which were not used in calibration, it scored better in 5 of 5 sets, although both forms underpredicted the strength of these tests at their highest stress.

Mode A: No Site Tests

Across all held-out tests (Table 2, Fig. 4a), the bounded form had a lower CRPS than the quadratic form, −0.0018 (95% CI −0.0034 to −0.0003; Wilcoxon p = 0.039, 22 of 36 publications). The 80% interval covered 84% of the tests and 9% fell below the 10th percentile. Above 2.75 MPa (100 tests, 9 publications), the bounded form reduced the mean standardized residual from +0.28 to +0.16, but the difference in CRPS was not significant (−0.0025 (95% CI −0.0056 to +0.0013; Wilcoxon p = 0.164, 7 of 9 publications)), and the predictive distribution was too wide (standard deviation of standardized residuals 0.74). The bounded form did not predict the strength of a new material better than Leps 2.0 with constant dispersion (−0.0027 (95% CI −0.0097 to +0.0030; Wilcoxon p = 0.727, 17 of 36 publications)); above 2.75 MPa the

reference had a lower 10% quantile loss. Below 0.5 MPa, however, the reference interval covered only 70% of tests, with the median biased low (mean standardized residual +0.59), whereas Leps-P covered 84%.

Mode B: Value of Site Tests

In Mode B (64 held-out tests from 13 publications), the prespecified level-only update reduced the CRPS from 0.0397 for the prior to 0.0278 (+0.0119 (95% CI +0.0060 to +0.0173; Wilcoxon $p = 0.013$, 1 of 13 publications) for prior minus update; Fig. 4b). It also scored better than a least-squares fit of ϕ_1' with m from the training data (0.0369; +0.0091 (95% CI +0.0003 to +0.0240; Wilcoxon $p = 0.040$, 2 of 13 publications)) and with $m = 0.90$ (0.0407, with 17% of tests below the 10th percentile). Joint updating of level and slope (0.0345) and discarding flagged sets before updating (0.0313) were not distinguishable from the primary rule. The level-only intervals were too wide (coverage 95% for a nominal 80%; standard deviation of standardized residuals 0.69). The conflict check flagged 7% of the Mode B sets, close to its nominal 5%.

Specimen Size

Sixteen materials of 12 publications were tested at maximum particle sizes differing by a factor of two or more. With set intercepts adjusted for void ratio, the between-set variance unrelated to size added to each estimate, and materials combined within publications, the random-effects mean of β_D was +0.035 (Knapp–Hartung 95% CI −0.012 to +0.082; Knapp and Hartung 2003), with a between-publication standard deviation of 0.049. A material-level analysis gave +0.017 (−0.024 to +0.059), and neither the change in uniformity coefficient nor angularity explained the heterogeneity (Fig. S1). A systematic size effect was therefore not detected; for

222 design, Leps-P uses a zero mean with this heterogeneity as uncertainty, which avoids crediting
223 the positive estimate that runs against Weibull scaling.
224 *Sources of Uncertainty at Field Scale*
225 For an untested material of low rock quality placed at $e = 0.65$ and scaled from 0.18 m to
226 1 m particles, the standard deviation of the mean-envelope ln τ at 3 MPa corresponds to about
227 3.5° in ϕ_s'. Specimen size contributed 47%, between-set variability 30%, publication 19%, the
228 unidentified asymptote 1%, and the fixed effects 2% (Fig. 5). At 10 MPa, beyond the data, the
229 asymptote contributed 4%. The extrapolation of the envelope form is therefore a smaller source
230 of uncertainty than the identity of the material and the scale of the particles.
231 **Application: Waste Rock from a Large Open-Pit Mine**
232 *Site Data*
233 The waste rock of a large open-pit mine hosted in carbonate rocks comprises five rock
234 types, A to E, of which A is a metacarbonate, B a silicate metasedimentary rock, C an igneous
235 rock, and D and E carbonate and argillaceous sedimentary rocks. Drained triaxial tests on large
236 specimens of scalped gradation by one laboratory were available for A, B, and C (5 sets).
237 Illustrative mass fractions of 35%, 25%, 20%, 10%, and 10% were used. Rock quality was taken
238 as low for A, C, D, and E and moderate for B. For rock type A, the conflict check gave $p = 0.16$
239 with the low class, 0.11 with the moderate class, and 0.025 with the good class, so the tests were
240 consistent with the class assigned.
241 *Conflict Check and Failure Criterion*
242 The conflict check flagged 3 of the 5 sets (Fig. 6a): both sets of rock type B ($p \leq 0.0006$)
243 and one set of rock type C ($p = 0.042$). In both B sets the secant angle increased with stress and
244 the specimens were still contracting at the maximum deviator, reached at 13–19% axial strain.

Because the full stress–strain curves were tabulated in the test reports, the sets were re-reduced with the deviator at 10% and 15% axial strain: the free log–log slope of the B sets remained above one (1.04–1.19), so the increase is not an artifact of the failure definition. Fixed-strain criteria created new conflicts in the other rock types, which confirms that site tests must be reduced with the criterion of the calibration database. Flagged sets did not update the envelope; because they are stronger than the prior at high stress, excluding them is conservative here, although not as a general rule. The remaining sets were updated jointly with a common laboratory effect.

Placement Density and Dump-Average Envelope

Laboratory specimens had void ratios of 0.31–0.42, whereas end-dumped waste rock is looser; a placement void ratio of 0.65 was adopted, with degradation of rock type E represented by an additional increase of 0.15. A failure surface through a mixed dump mobilizes the mass-weighted strength of the rock types it crosses; the envelopes were sampled jointly, sharing hyperparameters, fixed effects, laboratory effect, and size shift. At σ_n' = 3 MPa the median and 5th percentile of the dump-average ϕ_s' were 37.3° and 35.6° at laboratory density and size, and 33.2° and 28.8° for placed waste rock at field scale (Table 4; Fig. 6b). At 20 MPa the bounded form gave 28.9° and 23.4°, against 26.9° and 21.5° for the quadratic form. Full correlation between rock types lowered the 5th percentile at 3 MPa by 0.4°, and a surface confined to the weakest rock type (E) by 2.8°. The full-correlation case sorts the samples of each rock type at every stress, so it also removes the uncertainty of the slope.

Stability of High Dumps

Dumps 150–700 m high with an overall slope of 2H:1V on a competent, drained, level foundation were analyzed with the method of Bishop (1955) and the tabulated nonlinear

268 envelopes (Table 5). The unit weight was made consistent with the void ratio: 17.2 kN/m³ for e =
269 0.65 (specific gravity 2.75, water content 5%) and 21 kN/m³ for the laboratory density. For 700
270 m, the critical circle reached $\sigma_n' \approx 2.3$ MPa, within the calibration range. The factor of safety was
271 1.48 with the median and 1.25 with the 5th-percentile dump-average envelope, compared with
272 1.70 and 1.60 at laboratory density and specimen size; with a unit weight of 21 kN/m³ at e = 0.65
273 the latter values were 1.45 and 1.22. In a Monte Carlo analysis with 600 envelope samples per
274 correlation assumption, each sample was evaluated on a family of circles and, when the factor of
275 safety was below 1.3, by a full search of the critical surface. The model correlation gave 0
276 failures (probability of failure 0.0–0.6%, Clopper–Pearson 95% interval; Clopper and Pearson
277 1934), and a surface confined to the untested, degraded rock type E gave 11 (1.8%, 0.9–3.3%).
278 These are conditional probabilities that reflect strength uncertainty only, for a drained foundation
279 and a two-dimensional section. The factor of safety computed with the 5th-percentile envelope
280 was close to the 5th percentile of the factor of safety (1.25 versus 1.26), because shared level
281 shifts dominate the uncertainty here; this need not hold when pore pressure, curvature
282 uncertainty, or independent materials dominate.

283 **Discussion**

284 *What Leps-P Adds*

285 Leps-P does not predict the central strength of an untested material better than Leps 2.0.
286 Its contributions are an envelope whose curvature decays with stress, which removes the low
287 bias of fixed-form fits at high stress and is supported within materials; an uncertainty that is
288 calibrated at low stress, where the constant-dispersion reference is not, and wide but unbiased at
289 high stress; a prespecified updating rule that outperformed the simple fits used in practice; and a
290 decomposition showing that, for an untested waste rock, the material and the particle scale matter

more than the extrapolation of the envelope form. The last result indicates where investigation is most valuable: large-scale tests of the actual waste rock remove most of the material and laboratory components, and tests of one material at two particle sizes would constrain the largest remaining term.

Design Format

A design procedure consistent with these results is: (1) assign rock quality and placement void ratio to each rock type and check the conflict diagnostic for any site tests, reduced with the maximum-deviator criterion; (2) update with the tests that pass, at the level only (Eq. 7); (3) shift the envelopes to placement density and field particle size; (4) adopt as characteristic value for a global mechanism the 5th percentile of the mean envelope, a Bayesian cautious estimate of the mean in the sense of Eurocode 7 (CEN 2004, 2024; Prästings et al. 2019); (5) check the probability of failure by sampling the envelopes with stated correlations between rock types, including the weakest-rock-type scenario; and (6) choose the target factor of safety for the 5th-percentile envelope from the consequence class. Table 5 pairs both quantities for the example: a factor of safety of about 1.25 with the 5th-percentile envelope corresponded to no failures in 600 samples, and about 1.09 to a conditional probability of failure near 2%. Applying factors calibrated for best-estimate parameters (Read and Stacey 2009; Hawley and Cunning 2017) to the 5th-percentile envelope would count the strength uncertainty twice.

Limitations

The database contains no tests above σ_n' = 7.9 MPa and its tests above 3 MPa come from eight publications, so the bounded form is supported within the data and only its consequences are explored beyond; the asymptotic angle is not identified. The lower tail of the population is truncated at ϕ_1' = 31°, class labels may carry information from measured strength, and the

within-material residual is laboratory scatter rather than spatial variability in the dump. The site application uses one laboratory and illustrative proportions. Pore pressure in saturated basal layers, wetting collapse (Oldecop and Alonso 2001), seismic loading, three-dimensional geometry, and static liquefaction of contractive waste are outside the scope; they require their own analyses, for which the drained envelopes and their uncertainty are an input. Leps-P can be re-estimated as data appear, in particular drained tests at σ_3' of 5–10 MPa on carbonate and argillaceous waste rock and tests of one material at two sizes with parallel gradation.

Conclusions

1. The calibration data support an envelope whose curvature decays with stress toward an asymptote (p = 8e-05), consistent with the suppression of dilatancy; the asymptotic angle is not identified and is averaged over the likelihood ridge.
2. Out of sample, the bounded form scored better than a quadratic log–log form and removed the low bias of the median at high stress, but it did not predict the central strength of a new material better than Leps 2.0 with constant dispersion.
3. With a few site tests at low stress, a prespecified level-only update predicted the strength at higher stress better than least-squares fits with a fixed curvature; its intervals were wider than nominal.
4. No systematic specimen-size effect was detected within materials; the heterogeneity of the size effect is the largest source of uncertainty of an untested waste rock at field scale, followed by material and laboratory variability.
5. For a 700 m dump on a competent foundation, placement density and untested rock types controlled the margin of safety more than the form of the envelope.

Data Availability Statement

Some or all data, models, or code generated or used during the study are available in a repository or online in accordance with funder data retention policies (Leps-P code, parameters, and evaluation scripts: repository DOI; calibration database: Arrieta 2025). Some or all data, models, or code generated or used during the study are proprietary or confidential in nature and may only be provided with restrictions (the site test results, which may be provided in anonymized form with the permission of their owner).

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Supplemental Materials

Fig. S1 (specimen-size effect by material) and Table S1 (every Mode B procedure evaluated, for the bounded form and, in an earlier analysis, for the quadratic form with mixed updating rules) are available online in the ASCE Library (www.ascelibrary.org).

Notation

The following symbols are used in this paper:

$\mathbf{C}_\beta$ = covariance matrix of the fixed effects;

D_f, D_l = maximum particle size in the field and in the laboratory;

D^2 = prior predictive Mahalanobis distance of the site tests;

$\mathbf{G}_p$, $\mathbf{G}_s$ = covariance matrices of the publication and set random effects;

J, j = gradient of the mean envelope with respect to the fixed effects

r = vector of residuals of the site tests from the prior mean

S = *diagonal matrix of test-to-test variances*

$T = \tan \phi'_{cs}$;

Z, $\mathbf{z}$ = design rows (1, x*) of the random effects;

a_0, a_1 = parameters of the stress-dependent test-to-test dispersion;

$\mathbf{b}^{(p)}$, $\mathbf{b}^{(s)}$ = publication and set random effects on level and slope;

b_0, b_1 = level and slope deviations of a material in log–log space;

e = initial void ratio;

g_0 = prior variance of q_0;

m = exponent of the power envelope;

n = number of site tests;

p = p-value;

p_a = atmospheric pressure (0.1013 MPa);

q_0 = level of the envelope at $\overline{x}$;

r_i = residual of site test i;

$x = \ln(\sigma'_n/\sigma_0)$;

$\overline{x}$ = mean of x over the site tests of a set;

$y = \ln(\tau/\sigma_0)$;

β_c = coefficient of rock quality class c;

β_D = specimen-size coefficient;

β_e = effect of void ratio on η;

ε = test-to-test deviation;

ζ = decay rate of the dilatant part of strength;

$\eta = \ln \tan \phi'_1$ of the population;

382 σ_0 = reference stress, $10p_a$;

383 σ_3' = effective confining stress;

384 σ_n' = effective normal stress on the failure plane;

385 σ_ε = standard deviation of test-to-test scatter;

386 τ = shear strength;

387 ϕ_1' = secant friction angle at $\sigma_n' = \sigma_0$;

388 ϕ_{cs}' = asymptotic secant friction angle;

389 $\phi_{e=0}'$ = ϕ_1' extrapolated to $e = 0$ in Leps 2.0;

390 ϕ_s' = secant friction angle;

391 ω = decrease of ϕ_1' per unit void ratio in Leps 2.0;

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Figure Captions List

Fig. 1. Distribution of effective normal stress on the failure plane in the calibration database, compared with stresses in a 700 m waste dump.

Fig. 2. Design envelopes of Leps-P for a new material tested by a new laboratory, by rock quality and void ratio: median and 5th percentile of the mean envelope (characteristic value). The shaded zone exceeds the largest normal stress in the database.

Fig. 3. Evidence for stress-dependent curvature: (a) profile log-likelihood of the asymptotic angle and decay rate relative to its maximum (dot); and (b) difference in CRPS between the bounded and quadratic forms when each material is updated at low stress and predicted at higher stress, by publication (marker size increases with the number of tests; filled: bounded form better).

Fig. 4. Out-of-sample evaluation, leaving out each publication: (a) coverage of the 80% interval by range of normal stress without site tests; and (b) mean CRPS of ln τ for held-out tests at $\sigma_n' \geq$ 2 MPa with site tests at $\sigma_n' \leq 1.5$ MPa (bars: 95% bootstrap confidence intervals over publications; filled: prespecified procedure).

Fig. 5. Shares of the variance of the mean-envelope ln τ for an untested material of low rock quality placed at e = 0.65, scaled from 0.18 m to 1 m maximum particle size.

Fig. 6. Waste rock of the example: (a) laboratory-scale envelopes by rock type (thin lines: 5th percentile) with the site tests (open symbols: tests flagged by the conflict check); and (b) envelopes of placed waste rock (void ratio 0.65, maximum particle size 1 m) by rock type and dump-average envelope. Shaded zones are beyond the stresses tested at the site.

Table 1. Parameters of Leps-P (bounded form)

Parameter	Estimate	Standard error or 90% credible interval
β_c, good rock quality	0.223	0.034
β_c, moderate rock quality	0.098	0.028
β_c, low rock quality	0.023	0.024
β_e	−0.543	0.050
ϕ'_{cs} (°), weighted mean over the ridge	22.4	13–28 (ΔML ≤ 3)
ζ, weighted mean over the ridge	0.26	—
SD of level, publication	0.060	0.042–0.084
SD of slope, publication	0.041	0.029–0.058
Correlation of level and slope, publication	−0.42	−0.73 to 0.03
SD of level, set	0.082	0.074–0.091
SD of slope, set	0.032	0.025–0.038
Correlation of level and slope, set	−0.42	−0.59 to −0.22
σ_ε at $\sigma'_n = \sigma_0$	0.036	0.034–0.038
a_1	−0.302	−0.369 to −0.235

Note: 767 tests, 226 sets, 36 publications. Fixed effects at the maximum-likelihood point of the ridge (ϕ'_{cs} = 24°, ζ = 0.275); variance parameters: posterior median and 90% credible interval. Median ϕ'_1 at e = 0: 51.3°, 47.8°, and 45.7° for good, moderate, and low rock quality.

Table 2. Out-of-sample evaluation (leave one publication out)

Procedure	80% coverage	Below P10	CRPS
Mode A, all stresses: Leps-P bounded	84%	9%	0.0670
Mode A, all stresses: Leps-P quadratic	83%	9%	0.0688
Mode A, all stresses: Leps 2.0, constant dispersion	82%	6%	0.0696
Mode A, σ'_n > 2.75 MPa: Leps-P bounded	91%	4%	0.0443
Mode A, σ'_n > 2.75 MPa: Leps-P quadratic	88%	4%	0.0468
Mode A, σ'_n > 2.75 MPa: Leps 2.0, constant dispersion	96%	4%	0.0446
Mode B: prior, no site tests	97%	3%	0.0397
Mode B: fit of ϕ'_1, m = 0.90	81%	17%	0.0407
Mode B: fit of ϕ'_1, m from training data	81%	2%	0.0369
Mode B: joint update	92%	3%	0.0345

Mode B: level-only update after conflict check	92%	8%	0.0313
Mode B: level-only update (prespecified)	95%	5%	0.0278

Note: Mode A: 767 tests (36 publications); above 2.75 MPa, 100 tests (9 publications). Mode B: tests at $\sigma_n' \geq 2$ MPa predicted from tests at $\sigma_n' \leq 1.5$ MPa of the same set. Nominal values: 80% coverage and 10% below P10. CRPS in units of ln τ.

Table 3. Design values of the secant friction angle ϕ_s' (°) for an untested material: median / 5th percentile of the mean envelope

Rock quality	e	0.5 MPa	1 MPa	3 MPa	5 MPa	8 MPa
Good	0.30	50.1 / 44.4	46.8 / 41.8	42.0 / 37.3	40.0 / 34.9	38.3 / 32.8
	0.45	47.6 / 42.0	44.4 / 39.6	39.9 / 35.3	38.0 / 33.1	36.5 / 31.1
	0.65	44.3 / 38.8	41.4 / 36.6	37.2 / 32.8	35.6 / 30.8	34.2 / 29.0
Moderate	0.30	46.4 / 40.9	43.3 / 38.5	38.9 / 34.4	37.1 / 32.3	35.6 / 30.4
	0.45	43.9 / 38.5	41.0 / 36.3	36.9 / 32.5	35.3 / 30.6	33.9 / 28.8
	0.65	40.6 / 35.4	38.0 / 33.4	34.4 / 30.1	32.9 / 28.4	31.7 / 26.8
Low	0.30	44.1 / 38.7	41.2 / 36.5	37.1 / 32.7	35.4 / 30.7	34.0 / 28.9
	0.45	41.6 / 36.3	38.9 / 34.3	35.2 / 30.9	33.6 / 29.1	32.4 / 27.4
	0.65	38.4 / 33.3	36.0 / 31.5	32.7 / 28.5	31.4 / 27.0	30.3 / 25.5

Note: Laboratory specimen size, new laboratory, no site tests; bounded form averaged over the likelihood ridge. Values at 8 MPa are slightly beyond the database (7.9 MPa).

Table 4. Secant friction angle ϕ_s' (°) of the waste rock of the example, median / 5th percentile of the mean envelope

Material	Laboratory, 3 MPa	Placed, 1 MPa	Placed, 3 MPa	Placed, 10 MPa
Rock type A	36.5 / 35.1	35.5 / 31.2	32.3 / 28.3	29.6 / 24.6
Rock type B	38.9 / 35.7	38.4 / 33.1	34.7 / 29.7	31.5 / 26.0
Rock type C	36.8 / 34.6	37.0 / 31.7	33.6 / 28.6	30.6 / 25.1
Rock type D (untested)	36.9 / 33.0	36.4 / 30.4	33.0 / 27.6	30.1 / 24.2
Rock type E (untested)	36.9 / 33.0	34.1 / 28.3	31.2 / 25.9	28.8 / 23.0
Dump average, model correlation	37.3 / 35.6	36.5 / 31.9	33.2 / 28.8	30.3 / 25.3
Dump average, full correlation	37.3 / 34.8	36.5 / 31.4	33.1 / 28.4	30.2 / 24.8

Note: Laboratory: specimen void ratio and size. Placed: void ratio 0.65 (rock type E, 0.80), maximum particle size 1 m, size effect with zero mean. No site tests above $\sigma_n' \approx 4.6$ MPa.

Table 5. Factor of safety of dumps on a competent level foundation, placed waste rock (median / 5th-percentile envelope) and conditional probability of failure at 700 m

Envelope	150 m	400 m	700 m	Failures, 700 m (95% CI)
Dump average, model correlation	1.79 / 1.48	1.58 / 1.33	1.48 / 1.25	0/600 (0.0–0.6%)
Dump average, full correlation	1.79 / 1.44	1.58 / 1.30	1.48 / 1.23	0/600 (0.0–0.6%)
Weakest tested rock type only	1.72 / 1.41	1.52 / 1.29	1.43 / 1.22	0/600 (0.0–0.6%)
Untested, degraded rock type E only	1.61 / 1.24	1.44 / 1.15	1.37 / 1.09	11/600 (0.9–3.3%)

Note: Bishop method; unit weight 17.2 kN/m³. Monte Carlo with 600 envelope samples and a full search when FS < 1.3; Clopper–Pearson intervals. At laboratory density and specimen size (21 kN/m³) the 700 m dump gave 1.70 / 1.60 and 0 failures; with the quadratic form at e = 0.65, 1.47 / 1.24 and 0 failures.